

# SAIBALAJI NEELI

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## EDUCATION

### Oregon State University

Sep 2023 - Jun 2025

*Masters, Computer Science and Engineering (GPA: 3.52)*

- **Coursework:** Machine Learning, Machine Learning Challenges, Topological Data Analysis, Human Computer Interaction, Operating systems, Error Correcting Codes, Algorithms

### Anna University

Jan 2019 - Apr 2023

*B.E., Computer Science and Engineering (GPA: 3.60)*

- **Coursework:** Artificial Intelligence, Computer Architecture, Databases, Networks, Software engineering, Data warehouse

## WORK EXPERIENCE

### TDK Invensense | Machine learning intern - Capstone.

Sep 2024 - Jan 2025

- Developed a low-power people counting system using ultrasonic sensors and applied Data Analysis principles to enhance scalability and system efficiency.
- Designed and optimized ML models for embedded systems by leveraging ML Modeling techniques and statistical diagnostics to ensure accuracy in resource-constrained environments.
- Integrated hardware and software components to create a functional prototype, effectively utilizing Data Handling skills to generate actionable, data-driven insights.

### Supertutor AI | ML Development intern.

Jun 2024 - Sep 2024

- Developed machine learning models using Python, TensorFlow, and scikit-learn to enhance recommendation systems, applying Business Impact Measurement methods that increased user engagement by 25% and retention by 15%.
- Collaborated with cross-functional teams to identify key feature improvements and drive Framework Development initiatives that boosted app usability and contributed to a 50% rise in positive app store reviews.
- Executed complex SQL queries on a dataset of over 500,000 records and refined processes through Pipeline Building, uncovering actionable insights that improved engagement metrics by 15%.

### Outlier AI | Prompt Engineer.

Jan 2024 - Sep 2024

- Enhanced AI system functionality by debugging and optimizing response generation pipelines, incorporating Model Deployment techniques to ensure increased user satisfaction.
- Improved model accuracy and relevance by fine-tuning algorithms to better align outputs with organizational objectives.
- Collaborated with cross-functional teams to deploy AI solutions tailored to user needs, ensuring seamless integration and usability.

### Oregon State University | Graduate Research Assistant.

Jan 2024 - May 2024

- Conducted advanced research on minimal imputations in machine learning data preparation, resulting in a 15% reduction in data processing time.
- Evaluated over 10 machine learning models to optimize data completeness while maintaining high predictive accuracy, improving overall efficiency by 25%.
- Co-authored a research paper for VLDB 2024, demonstrating expertise in data preprocessing, machine learning, and statistical analysis.

## PROJECTS

### Life Expectancy Prediction

Jan 2024 - Mar 2024

- Developed a regression model to predict life expectancy using a dataset of 200,000+ records spanning 15 years (2000-2015).
- Achieved a high model performance with an  $R^2$  value of 0.85.
- Preprocessed the dataset to handle missing values, outliers, and inconsistencies to enhance model accuracy.
- Utilized Python and scikit-learn for data analysis and model development.

### Pneumonia Detection and Classification

Apr 2023 - Jul 2023

- Implemented Convolutional Neural Networks (CNNs) for pneumonia detection using chest X-ray images.
- Achieved 95% classification accuracy and 0.92 ROC-AUC, surpassing the baseline models.
- Incorporated Transfer Learning with pre-trained models for improved diagnostic performance.
- Leveraged TensorFlow and Keras to streamline model training and testing processes.

### Database Management System

Jan 2023 - Mar 2023

- Designed and implemented a centralized SQL database to manage F-1 race data, consisting of over 10,000 race statistics.
- Optimized database queries, reducing data retrieval time by 30%.
- Applied indexing and normalization techniques to improve system performance.
- Utilized MySQL and SQL Server for database creation, management, and analysis.

## SKILLS

- **Programming Languages:** C, C++, Python, JavaScript, Java, HTML, CSS, R programming, SQL
- **Databases:** MongoDB, MySQL, PostgreSQL
- **Tools & Platforms:** Microsoft Excel, Tableau, PowerBI, Jupyter, Google colab, Matplotlib
- **Frameworks & Libraries:** Pandas, NumPy, TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, Hadoop
- **Operating Systems:** Windows, Linux, MacOS
- **Machine Learning:** Supervised and Unsupervised Learning, Neural Networks, Deep Learning, Reinforcement Learning, NLP
- **Data Science & Operational Expertise:** Data Science, Building pipelines, Operational recommendations, Healthcare concepts, Model performance measurement